

## FORT KNOX, KY, USA

WMO: 724240

Lat: 37.900N

Lon: 85.967W

Elev: 755

StdP: 14.30

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13807

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 7.0        | 13.2 | -0.9                          | 5.4 | 10.7 | 4.1 | 7.0 | 15.5 | 22.8                  | 46.7 | 20.5 | 43.1 | 6.1                   | 280  | 0.438 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.6                   | 92.4            | 75.7 | 90.1 | 75.4 | 88.0 | 74.4 | 78.9                | 88.1 | 77.6 | 86.5 | 76.3 | 84.7 | 7.6                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.5                            | 142.2 | 83.6 | 75.0 | 135.1 | 82.3 | 73.5 | 128.2 | 80.4 | 42.9          | 88.1 | 41.5 | 86.5 | 40.2 | 84.8 | 87.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 19.4          | 17.1 | 15.3 | -0.3                       | 95.4 | 7.6                | 3.9 | -5.8                                               | 98.1 | -10.3      | 100.4 | -14.6      | 102.6 | -20.2      | 105.4 |    |    |
| (5)               |      |      | -0.9                       | 81.1 | 7.5                | 2.1 | -6.3                                               | 82.6 | -10.7      | 83.8  | -14.8      | 85.0  | -20.3      | 86.6  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 56.5   | 33.0  | 38.2  | 46.1  | 56.9  | 66.0  | 73.6  | 76.3  | 75.7  | 69.2  | 58.3  | 46.4  | 37.6  |  |  |
|                                                                        | DBStd       | 17.60  | 12.36 | 11.69 | 10.87 | 9.21  | 7.71  | 5.49  | 4.77  | 4.82  | 7.04  | 8.73  | 9.70  | 10.62 |  |  |
|                                                                        | HDD50       | 1755   | 545   | 358   | 204   | 38    | 1     | 0     | 0     | 0     | 0     | 24    | 180   | 405   |  |  |
|                                                                        | HDD65       | 4392   | 993   | 751   | 590   | 270   | 87    | 8     | 1     | 1     | 39    | 244   | 559   | 849   |  |  |
|                                                                        | CDD50       | 4143   | 17    | 27    | 85    | 246   | 498   | 707   | 816   | 796   | 576   | 281   | 73    | 21    |  |  |
|                                                                        | CDD65       | 1300   | 0     | 1     | 5     | 27    | 118   | 265   | 352   | 331   | 164   | 35    | 2     | 0     |  |  |
|                                                                        | CDH74       | 11599  | 0     | 2     | 41    | 237   | 981   | 2357  | 3219  | 2987  | 1453  | 310   | 12    | 0     |  |  |
|                                                                        | CDH80       | 4197   | 0     | 0     | 3     | 32    | 250   | 848   | 1285  | 1183  | 528   | 67    | 1     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 6.7    | 8.2   | 8.0   | 8.1   | 8.2   | 6.5   | 5.6   | 5.0   | 4.8   | 5.1   | 6.0   | 7.0   | 7.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 48.3   | 3.6   | 3.3   | 4.6   | 4.8   | 5.5   | 4.2   | 4.3   | 3.3   | 3.0   | 3.6   | 3.4   | 4.4   |  |  |
|                                                                        | PrecMax     | 71.4   | 6.3   | 7.2   | 14.8  | 17.4  | 11.3  | 9.3   | 7.6   | 7.2   | 10.0  | 9.0   | 8.1   | 10.9  |  |  |
|                                                                        | PrecMin     | 36.4   | 1.0   | 1.3   | 1.7   | 1.3   | 1.8   | 0.7   | 0.6   | 0.9   | 0.2   | 0.0   | 0.9   | 1.5   |  |  |
|                                                                        | PrecStd     | 7.8    | 1.4   | 1.5   | 2.6   | 2.9   | 2.2   | 2.0   | 1.7   | 1.7   | 2.3   | 2.3   | 1.7   | 1.9   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 66.0  | 71.1  | 79.1  | 83.5  | 87.9  | 92.2  | 97.7  | 96.2  | 94.3  | 86.1  | 75.3  | 68.1  |  |  |
|                                                                        |             | MCWB   | 60.1  | 59.5  | 64.2  | 65.5  | 71.9  | 76.5  | 76.7  | 76.6  | 70.9  | 70.5  | 62.7  | 61.6  |  |  |
|                                                                        | 2%          | DB     | 62.1  | 65.7  | 73.2  | 80.5  | 85.2  | 90.1  | 92.0  | 91.7  | 89.9  | 81.3  | 70.2  | 63.1  |  |  |
|                                                                        |             | MCWB   | 58.2  | 57.4  | 59.7  | 64.4  | 70.2  | 75.6  | 77.5  | 75.3  | 72.6  | 66.9  | 59.8  | 58.5  |  |  |
|                                                                        | 5%          | DB     | 57.4  | 61.6  | 68.0  | 76.6  | 83.1  | 87.8  | 89.6  | 89.3  | 86.2  | 77.2  | 65.8  | 58.5  |  |  |
|                                                                        |             | MCWB   | 53.6  | 54.8  | 56.7  | 62.3  | 69.3  | 74.4  | 76.8  | 74.9  | 71.2  | 64.5  | 58.2  | 54.2  |  |  |
|                                                                        | 10%         | DB     | 52.4  | 56.8  | 63.3  | 72.7  | 80.6  | 85.5  | 87.0  | 86.5  | 82.4  | 72.9  | 62.3  | 54.3  |  |  |
|                                                                        |             | MCWB   | 48.1  | 50.7  | 54.1  | 60.6  | 68.5  | 73.3  | 75.3  | 73.7  | 69.5  | 62.7  | 55.9  | 50.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 62.2  | 63.0  | 66.8  | 69.7  | 75.2  | 79.7  | 81.7  | 80.3  | 76.6  | 73.2  | 66.0  | 63.2  |  |  |
|                                                                        |             | MCDB   | 64.3  | 67.8  | 76.3  | 78.3  | 83.6  | 89.0  | 89.3  | 90.2  | 86.8  | 83.0  | 71.3  | 66.7  |  |  |
|                                                                        | 2%          | WB     | 58.8  | 59.5  | 62.5  | 67.5  | 73.1  | 77.7  | 79.3  | 78.2  | 75.0  | 70.2  | 62.9  | 59.3  |  |  |
|                                                                        |             | MCDB   | 61.7  | 63.8  | 69.1  | 75.7  | 81.8  | 86.6  | 88.4  | 87.6  | 84.5  | 76.3  | 66.8  | 62.3  |  |  |
|                                                                        | 5%          | WB     | 54.7  | 56.3  | 59.3  | 65.1  | 71.5  | 76.2  | 78.0  | 76.9  | 73.5  | 67.4  | 59.8  | 55.0  |  |  |
|                                                                        |             | MCDB   | 57.6  | 60.1  | 65.5  | 72.4  | 80.1  | 84.8  | 86.8  | 85.8  | 81.6  | 74.2  | 64.9  | 57.4  |  |  |
|                                                                        | 10%         | WB     | 48.6  | 51.2  | 55.8  | 62.7  | 69.8  | 74.7  | 76.7  | 75.4  | 72.0  | 64.7  | 56.2  | 50.7  |  |  |
|                                                                        |             | MCDB   | 51.9  | 55.6  | 62.1  | 69.9  | 76.9  | 82.9  | 84.8  | 83.6  | 79.4  | 70.8  | 61.5  | 53.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.2  | 18.0  | 19.6  | 21.7  | 19.7  | 19.2  | 18.6  | 19.8  | 21.2  | 21.5  | 18.6  | 15.5  |  |  |
|                                                                        |             | MCDBR  | 19.3  | 23.9  | 24.9  | 24.6  | 22.6  | 21.8  | 21.4  | 22.2  | 25.0  | 24.8  | 22.7  | 19.6  |  |  |
|                                                                        |             | MCWBR  | 15.9  | 18.1  | 15.1  | 13.2  | 10.4  | 9.5   | 8.6   | 8.7   | 10.0  | 12.4  | 15.1  | 16.1  |  |  |
|                                                                        |             | MCDBR  | 17.6  | 21.6  | 21.3  | 20.1  | 19.1  | 19.0  | 19.3  | 19.0  | 20.0  | 20.2  | 19.4  | 17.9  |  |  |
|                                                                        | 5% WB       | MCWBR  | 15.5  | 18.7  | 15.3  | 13.2  | 9.9   | 9.3   | 8.9   | 8.4   | 9.3   | 11.7  | 15.2  | 16.5  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.310 | 0.323 | 0.354 | 0.403 | 0.441 | 0.455 | 0.481 | 0.484 | 0.419 | 0.359 | 0.329 | 0.306 |  |  |
|                                                                        | taud        |        | 2.475 | 2.413 | 2.394 | 2.255 | 2.189 | 2.183 | 2.122 | 2.104 | 2.266 | 2.435 | 2.470 | 2.493 |  |  |
|                                                                        | Ebn at Noon |        | 278   | 288   | 288   | 278   | 268   | 263   | 255   | 251   | 263   | 271   | 267   | 271   |  |  |
|                                                                        | Edh at Noon |        | 26    | 31    | 35    | 43    | 46    | 47    | 49    | 49    | 39    | 30    | 26    | 24    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 620   | 836   | 1180  | 1566  | 1747  | 1980  | 1879  | 1752  | 1450  | 1085  | 746   | 533   |  |  |
|                                                                        | RadStd      |        | 67    | 114   | 112   | 111   | 159   | 112   | 113   | 101   | 133   | 119   | 81    | 53    |  |  |

## Historical Trends

|                              |     | DBAvg  |        | Heating |       | Cooling |       | Degree-Days |       |       |  |
|------------------------------|-----|--------|--------|---------|-------|---------|-------|-------------|-------|-------|--|
|                              |     | 99% DB | 99% DP | 1% DB   | 1% WB | 1% DP   | HDD50 | HDD65       | CDD50 | CDD65 |  |
| (d)                          | (e) | (f)    | (g)    | (h)     | (i)   | (j)     | (k)   | (l)         | (m)   |       |  |
| (46) Station Only            | N/A | N/A    | N/A    | N/A     | N/A   | N/A     | N/A   | N/A         | N/A   | N/A   |  |
| (47) Regional (28 neighbors) | N/A | N/A    | N/A    | N/A     | N/A   | N/A     | N/A   | N/A         | +212  | +140  |  |

Nomenclature: See separate page